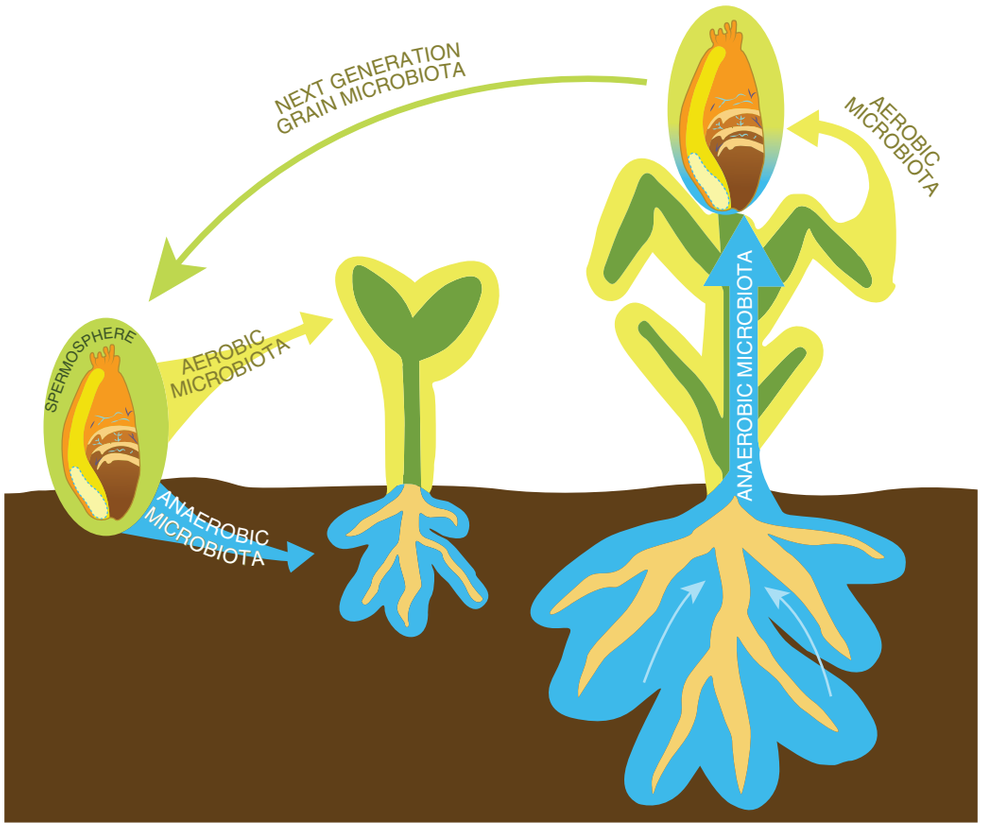



**Figure 10.8:** Spermosphere

## 10.8 Spermosphere

The region which is adjacent to the seed surface is termed as spermosphere (Figure 10.8). Healthy seeds carry specific bacterial flora in respect to number and species. There are several reports in the literature on the quantity and quality of microorganisms carried by the seeds of different plants species both externally and internally. When the seed is sown in soil, certain interactions between the seed borne microflora and the soil microorganisms take place under the influence of chemicals excluded by the germinating seed.

## Summary

Soil is the outer most covering of the earth. Soil consists of living and non living components that contribute its fertility. There are five major components in the soil, that includes mineral matter, water, air organic matter and living organisms. The soil environment is unique in several ways. It consists of bacteria, fungi, actinomycetes, algae and protozoa. Several factors influences the moisture, pH, temperature, organic and inorganic matter of the soil.

Microorganisms in soil interact themselves and lead to both beneficial and harmful interactions. Beneficial interaction includes symbiosis and commensalism. Harmful microbial interaction includes parasitism. The region adjacent to the root system is called rhizosphere. Bacteria predominate in rhizophere. Soil and their growth is influenced by nutritional substances released from plant tissues.



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